ENGINEERING REBOOT 2026

Week 03 – Day 11 Field Manual

Lists and Collections

Version 1.2 – Re-Baselined Edition – Publication Edition

Status: LOCKED

PHASE 0 – MISSION BRIEF

Objective

Learn how software systems store multiple related values using collections. Day 11 introduces the first true data structure: The List.

Core Concepts

```
• Lists
• Indexing
• len()
• Collection Storage
• Collection Access
• Collection Analysis
```

Engineering Applications

```
Inventory Systems
inventory = ["Laptop", "Monitor", "Keyboard"]

Sensor Readings
temperatures = [71,73,75,72,70]

Customer Lists
customers = ["Alice","Bob","Charlie"]

Automation Tasks
servers = ["web01","web02","db01"]
```

Learning Objectives

```
• Create lists
• Access list elements
• Use indexing
• Determine list length
• Process simple collections
• Apply collection concepts to practical software problems
```

Deliverable Inventory

```
1. list_basics.py
2. shopping_list.py
3. list_statistics.py
4. Notes.md
5. Journal.md
6. Evidence Package
```

PHASE 1 – LIST BASICS

Filename

```
list_basics.py
```

Implement

```
fruits = ["Apple", "Orange", "Banana", "Mango", "Grape"]

Display:
• Entire list
• First item
• Last item
```

Validation Matrix

```
['Apple', 'Orange', 'Banana', 'Mango', 'Grape']

First Item: Apple

Last Item: Grape
```

Evidence Checkpoint

```
list_basics-Code.png

list_basics-Output.png
```

Completion Checkpoint

■ Exercise Completed

■ Validation Completed

■ Evidence Captured

PHASE 2 – SHOPPING LIST

Filename

```
shopping_list.py
```

Implement

```
shopping_list = [
    "Milk",
    "Bread",
    "Eggs",
    "Cheese"
]

Display:
• Entire shopping list
• Total item count

Use:
len()
```

Validation Matrix

```
Shopping List

Milk
Bread
Eggs
```

Cheese

Total Items: 4

Evidence Checkpoint

shopping_list-Code.png

shopping_list-Output.png

Completion Checkpoint

■ Exercise Completed

■ Validation Completed

■ Evidence Captured

PHASE 3 – LIST STATISTICS

Filename

list_statistics.py

Implement

```
numbers = [10, 20, 30, 40, 50]

Display:
• Sum
• Maximum
• Minimum
• Length

Use:
sum()
max()
min()
len()
```

Validation Matrix

Sum: 150

Maximum: 50

Minimum: 10

Length: 5

Evidence Checkpoint

list_statistics-Code.png

list_statistics-Output.png

Completion Checkpoint

■ Exercise Completed

■ Validation Completed

■ Evidence Captured

PHASE 4 – NOTES

```
03-Week-03
■■■ Day-11
    ■■■ Notes
        ■■■ Notes.md
```

Required Sections

- Concepts Learned
- Key Commands
- Key Observations
- Lessons Learned
- Questions For Future Study

Template

```
# Concepts Learned

# Key Commands

# Key Observations

# Lessons Learned

# Questions For Future Study
```

■ Notes Completed

PHASE 5 – JOURNAL

```
03-Week-03
■■■ Day-11
    ■■■ Journal
        ■■■ Journal.md
```

Required Sections

- Summary
- Challenges Encountered
- Solutions Applied
- Confidence Assessment
- Reflection
- Tomorrow Preparation

Suggested Reflection

How is a list different from storing values in separate variables?

Template

```
# Summary

# Challenges Encountered

# Solutions Applied

# Confidence Assessment

# Reflection

# Tomorrow Preparation
```

■ Journal Completed

PHASE 6 – GIT OPERATIONS

```
git status

git add .

git commit -m "Complete Day 11 list exercises"

git push

git fetch
git status
```

git branch -vv

Preferred End State

Your branch is up to date with 'origin/main'

nothing to commit, working tree clean

Evidence Checkpoint

Git-Operations.png

Must Show

```
git add .
git commit
git push
git fetch
git status
git branch -vv
```

■ Git Operations Completed

■ Repository Synchronized

■ Evidence Captured

END-OF-DAY SUCCESS CRITERIA

■ list_basics.py completed
■ shopping_list.py completed
■ list_statistics.py completed
■ All validation completed
■ Exercise evidence captured
■ Notes completed
■ Journal completed
■ Git workflow completed
■ Git-Operations.png captured
■ Remote synchronization verified
■ Working tree clean

MISSION COMPLETE ✓

DAY 12 PREVIEW

Topic

Collection Traversal & Iteration

Expected Deliverables

• iterate_names.py
• number_list.py
• grade_report.py

Core Concepts

• for loops
• traversal
• iteration
• counters
• collection processing

Engineering Focus

Processing datasets rather than individual values.